3. Types of AI (Based on Capabilities)

3.1 Artificial Narrow AI (Weak AI) – Present-day Applications

Artificial Narrow Intelligence, also known as Weak AI (what we refer to as Narrow AI), is the only type of AI that exists today. Any other form of AI is theoretical. It can be trained to perform a single or narrow task, often far faster and better than a human mind can.

However, it can't perform outside of its defined task. Instead, it targets a single subset of cognitive abilities and advances in that spectrum. Siri, Amazon's Alexa and IBM Watson® are examples of Narrow AI. Even OpenAI's ChatGPT is considered a form of Narrow AI because it's limited to the single task of text-based chat.

3.2 Artificial General AI (Strong AI) – Theoretical Concepts

Artificial General Intelligence (AGI), also known as Strong AI, is today nothing more than a theoretical concept. AGI can use previous learnings and skills to accomplish new tasks in a different context without the need for human beings to train the underlying models. This ability allows AGI to learn and perform any intellectual task that a human being can.

3.3 Super AI – Beyond Human Cognition

Super AI is commonly referred to as artificial superintelligence and, like AGI, is strictly theoretical. If ever realized, Super AI would think, reason, learn, make judgements and possess cognitive abilities that surpass those of human beings.

The applications possessing Super AI capabilities will have evolved beyond the point of understanding human sentiments and experiences to feel emotions, have needs and possess beliefs and desires of their own.

Figure 2. The three kinds of AI based on capabilities

IBM Data and AI Team, 2024

Source: https://www.ibm.com/think/topics/artificial-intelligence-types

Example Questions: Topic 3

Types of AI (Based on Capabilities) — 10 questions

1. Which type of AI is described in the study guide as the only type that exists today?
   A. Super AI
   B. Artificial General AI
   C. Artificial Narrow AI
   D. Artificial Superintelligence

2. Artificial Narrow AI is also known as:
   A. Strong AI
   B. Weak AI
   C. Super AI
   D. Human-level AI

3. What is a key limitation of Artificial Narrow AI?
   A. It cannot process any data
   B. It cannot perform outside its defined task
   C. It always performs every intellectual task
   D. It surpasses human cognition

4. Which of the following is listed as an example of Narrow AI in the study guide?
   A. A theoretical AGI system
   B. Super AI
   C. Amazon’s Alexa
   D. A machine with cognition beyond humans

5. Why does the study guide classify ChatGPT as Narrow AI?
   A. It is stated to be limited to text-based chat
   B. It can perform any intellectual task a human can
   C. It is beyond human cognition
   D. It has beliefs and desires of its own

6. Artificial General Intelligence (AGI) is also known as:
   A. Weak AI
   B. Strong AI
   C. Narrow AI
   D. Artificial Narrow Intelligence

7. According to the study guide, AGI could use previous learnings and skills to:

A. Complete only one fixed task

B. Accomplish new tasks in a different context

C. Avoid learning new tasks

D. Perform only text-based chat

8. Which statement about AGI matches the study guide?

A. It exists widely today

B. It is strictly a present-day application

C. It is a theoretical concept

D. It cannot transfer skills to new contexts

9. What would distinguish Super AI from human intelligence, according to the study guide?

A. It would only sort data

B. It would possess cognitive abilities that surpass those of humans

C. It would be limited to a single narrow task

D. It would require humans to train every underlying model for each new task

10. Which sequence correctly moves from the most limited capability to the greatest capability?

A. Super AI → AGI → Narrow AI

B. AGI → Narrow AI → Super AI

C. Narrow AI → AGI → Super AI

D. Narrow AI → Super AI → AGI

Answer Key

1. C — Artificial Narrow AI
2. B — Weak AI
3. B — It cannot perform outside its defined task
4. C — Amazon’s Alexa
5. A — It is stated to be limited to text-based chat
6. B — Strong AI
7. B — Accomplish new tasks in a different context
8. C — It is a theoretical concept
9. B — It would possess cognitive abilities that surpass those of humans
10. C — Narrow AI → AGI → Super AI